Department of Science & Humanities

AI-Powered Defence Cyber Incident & Safety Portal

- Threat Detection
- Risk Classification
- CERT Escalation

Presented at YUDHISTRA 2026 – Inter-College Innovation Competition

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CERTIFICATE OF AUTHENTICITY

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This is to certify that the research paper titled:

"RAKSHAK AI – AI-Powered Defence Cyber Incident & Safety Portal"

has been presented as part of YUDHISTRA 2026 – Inter College Innovation Competition, based on Cybersecurity Problem Statement Reference ID: 25183 (inspired by SIH 2025), by the following students of the Department of Information Technology, Rathinam Technical Campus. The work presented in this document is authentic and carried out under proper academic supervision.

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The rapid increase in cyber threats targeting defence personnel, veterans, and their families has created a critical need for a dedicated cyber incident reporting and analysis platform. Existing public cybercrime reporting systems often handle a large volume of civilian complaints, resulting in delayed identification and response to defence-related cyber threats such as phishing attacks, fraudulent communications, malicious links, identity theft, social engineering, and operational security risks. To address these challenges, this paper presents Rakshak AI, an AI-enabled Defence Cyber Incident and Safety Portal designed to provide secure complaint registration, intelligent threat analysis, risk-based classification, and structured cyber incident management.

The proposed system enables users to submit cyber complaints along with digital evidence including suspicious text messages, URLs, screenshots, images, and files for further investigation. The platform incorporates Artificial Intelligence and Machine Learning techniques to analyze submitted content and detect potential cyber threats using phishing pattern recognition, Natural Language Processing (NLP), keyword analysis, and risk scoring mechanisms. Based on the analysis, incidents are categorized into multiple severity levels such as Low, Medium, High, and Critical to support faster triage and prioritization.

The system is developed using modern web technologies including React, Firebase Authentication, Firestore Database, and responsive frontend frameworks to ensure secure access, efficient complaint handling, and user-friendly interaction. The portal also includes administrative dashboards for monitoring incidents, viewing analytics, maintaining audit records, and managing cyber complaint workflows. In cases where complaints are classified as Critical severity, the system automatically escalates the incident and sends real-time email alerts to CERT administrators for immediate attention and response. Experimental testing demonstrates that the proposed system improves cyber threat visibility, enhances complaint management efficiency, and supports proactive cyber safety awareness for defence communities. Rakshak AI aims to provide an intelligent and scalable cyber safety solution capable of strengthening digital defence preparedness through AI-assisted threat detection and centralized cyber incident management.

Index Terms — *Cybersecurity, Artificial Intelligence, Machine Learning, Natural Language Processing, Phishing Detection, Defence Technology, Threat Intelligence, Incident Management, CERT Escalation, Risk Classification*

Research Domain: Cybersecurity, Artificial Intelligence, Threat Intelligence, Defence Technology

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KEY CONTRIBUTIONS

- AI-powered phishing and scam detection
- Automated CERT escalation mechanism
- Defence-oriented cyber incident reporting portal
- Hybrid NLP + Machine Learning threat analysis
- Real-time dashboard monitoring and analytics
- Seeded cybersecurity research dataset generation
- Role-based secure cyber complaint management
- AI-assisted risk classification and prioritization

1. INTRODUCTION

With the rapid growth of digital communication, cyber threats targeting defence personnel, veterans, and their families have increased significantly. Phishing attacks, fake military-related messages, malicious links, identity theft, and social engineering attacks can lead to serious security and operational risks. Existing public cybercrime reporting systems often handle a large number of civilian complaints, resulting in delayed response and limited priority for defence-related incidents.

To address this challenge, Rakshak AI is proposed as an AI-powered Defence Cyber Incident and Safety Portal that enables secure cyber complaint reporting, intelligent threat analysis, and risk-based incident classification. The system uses Artificial Intelligence and Machine Learning techniques to analyze suspicious URLs, scam messages, screenshots, and other digital evidence to identify potential cyber threats. The portal also provides administrative dashboards, complaint tracking, risk monitoring, and cyber safety support to improve incident management and strengthen digital defence preparedness.

2. PROBLEM STATEMENT

In recent years, cyber threats targeting defence personnel, veterans, and their families have increased rapidly through phishing attacks, malicious links, fake military communications, identity theft, and social engineering activities. These individuals are considered high-value targets, where compromised personal data or digital communication may lead to serious national security and operational risks.

Currently, cyber complaints are mainly handled through the National Cyber Crime Reporting Portal (NCRP), which processes a large volume of civilian cases without prioritizing defence-related incidents. As a result, critical cyber threats linked to defence communities may experience delayed detection, analysis, and response.

To address this issue, there is a need for a dedicated AI-enabled Defence Cyber Incident and Safety Portal capable of securely collecting cyber complaints, analyzing suspicious content using AI/ML techniques, generating real-time threat alerts, and supporting risk-based incident management. The system should also assist administrators with structured dashboards, complaint monitoring, and prioritized cyber threat handling to improve digital defence preparedness and cyber safety awareness.

3. EXISTING SYSTEM

At present, cybercrime complaints in India are primarily handled through the National Cyber Crime Reporting Portal (NCRP), which is designed for the general public. Although the portal supports reporting of various cyber incidents, it does not provide a dedicated mechanism for defence personnel, veterans, or their families. Due to the large number of civilian complaints, defence-related cyber threats may not receive immediate attention or priority handling.

The existing system mainly focuses on manual complaint processing and lacks advanced AI-based threat analysis capabilities for identifying phishing attacks, malicious URLs, espionage patterns, and targeted cyber campaigns. In addition, there is limited support for real-time risk classification, automated mitigation guidance, and defence-specific cyber incident management.

Furthermore, existing platforms often lack centralized dashboards for prioritized monitoring, structured threat escalation, and efficient handling of high-risk incidents related to national security concerns. These limitations highlight the need for a dedicated AI-powered cyber safety portal specifically designed for defence communities.

3.1 Limitations of Existing System

- No defence-specific cyber complaint management system
- Limited AI-based cyber threat analysis capability
- No automated risk scoring mechanism
- Lack of real-time threat alert generation
- No centralized CERT escalation workflow
- Limited phishing URL and scam detection
- Lack of structured dashboard monitoring
- No AI-assisted threat prioritization
- Limited audit trail and incident intelligence support

3.2 Comparative Analysis

Feature	Existing NCRP Portal	Rakshak AI
Defence-Specific Cyber Complaint Support	No	Yes
AI-Based Threat Analysis	Limited	Yes
Automated Risk Classification	Limited	Yes
Real-Time Threat Alerts	No	Yes
CERT Escalation Mechanism	No	Yes
Phishing URL Detection	Limited	Yes
Dashboard-Based Incident Monitoring	Limited	Yes
Role-Based Access Control (RBAC)	Limited	Yes
AI-Assisted Threat Prioritization	No	Yes
Centralized Defence Cyber Monitoring	No	Yes

The comparative analysis demonstrates that the proposed Rakshak AI system provides several advanced capabilities beyond existing public cybercrime reporting platforms. Unlike traditional systems, Rakshak AI integrates AI-assisted threat analysis, automated risk classification, real-time CERT escalation, and defence-oriented cyber incident monitoring to improve cyber threat detection and response efficiency.

3.3 Novelty of the Proposed System

The proposed Rakshak AI system introduces a defence-oriented AI-assisted cyber incident management platform specifically designed for defence personnel, veterans, and their families. Unlike traditional cybercrime portals, the proposed system integrates hybrid AI-based threat analysis, phishing detection, risk classification, automated CERT escalation, and centralized dashboard monitoring into a unified architecture.

The novelty of the system lies in combining Artificial Intelligence, Natural Language Processing (NLP), phishing intelligence analysis, real-time risk scoring, and defence-specific cyber complaint prioritization within a scalable incident management framework. Additionally, the integration of seeded cyber threat datasets, simulated phishing campaigns, automated email escalation, and evidence intelligence handling improves practical applicability for cybersecurity research and cyber defence preparedness.

4. PROPOSED SYSTEM

The proposed system, Rakshak AI, is an AI-powered Defence Cyber Incident and Safety Portal designed to provide secure cyber complaint reporting, intelligent threat analysis, and efficient incident management for defence personnel, veterans, and their families. The system enables users to submit suspicious URLs, scam messages, screenshots, images, and other digital evidence through a secure and user-friendly web portal.

The portal integrates Artificial Intelligence and Machine Learning techniques to analyze submitted content and identify potential cyber threats such as phishing attacks, malicious links, fraud attempts, identity compromise, and social engineering activities. Based on the analysis, the system classifies incidents into multiple risk levels including Low, Medium, High, and Critical for faster triage and prioritization.

The proposed system also provides real-time alerts and cyber safety recommendations to help users respond effectively to detected threats. In addition, an administrative dashboard is included for monitoring complaints, viewing analytics, managing incidents, and maintaining audit records. The system aims to improve cyber threat visibility, strengthen digital defence preparedness, and support proactive cyber safety awareness through AI-assisted technologies. Additionally, when the system detects complaints categorized under Critical risk level, an automated escalation mechanism is triggered to notify CERT administrators through real-time email alerts. This enables faster incident response, priority handling, and improved cyber threat management for defence-related security incidents.

5. SYSTEM ARCHITECTURE

The architecture of Rakshak AI is designed to provide secure cyber complaint handling, intelligent threat analysis, and efficient incident management for defence communities. The system follows a modular architecture consisting of User Interface Layer, Authentication Module, AI Threat Analysis Engine, Database Layer, and Administrative Dashboard.

Initially, users access the web portal through a secure authentication system and submit cyber complaints along with digital evidence such as suspicious URLs, scam messages, screenshots, images, or files. The submitted data is then processed by the AI Threat Analysis Engine, where Artificial Intelligence and Machine Learning techniques analyze the content for phishing patterns, malicious activities, scam indicators, and potential cyber threats.

After analysis, the system performs risk classification and categorizes incidents into Low, Medium, High, or Critical severity levels. Based on the risk score, the portal generates real-time alerts and mitigation recommendations for users. If the detected threat is classified as Critical, the system automatically initiates an escalation workflow and sends email notifications to CERT administrators containing complaint details, risk level, threat type, and AI-generated analysis results for immediate review and action. All complaint details, analysis results, and activity records are securely stored in the database for monitoring and tracking purposes.

The Administrative Dashboard allows authorized administrators to monitor incidents, view analytics, manage complaints, maintain audit records, and prioritize high-risk cyber threats. This architecture ensures secure communication, efficient cyber threat detection, centralized incident management, and improved digital defence preparedness.

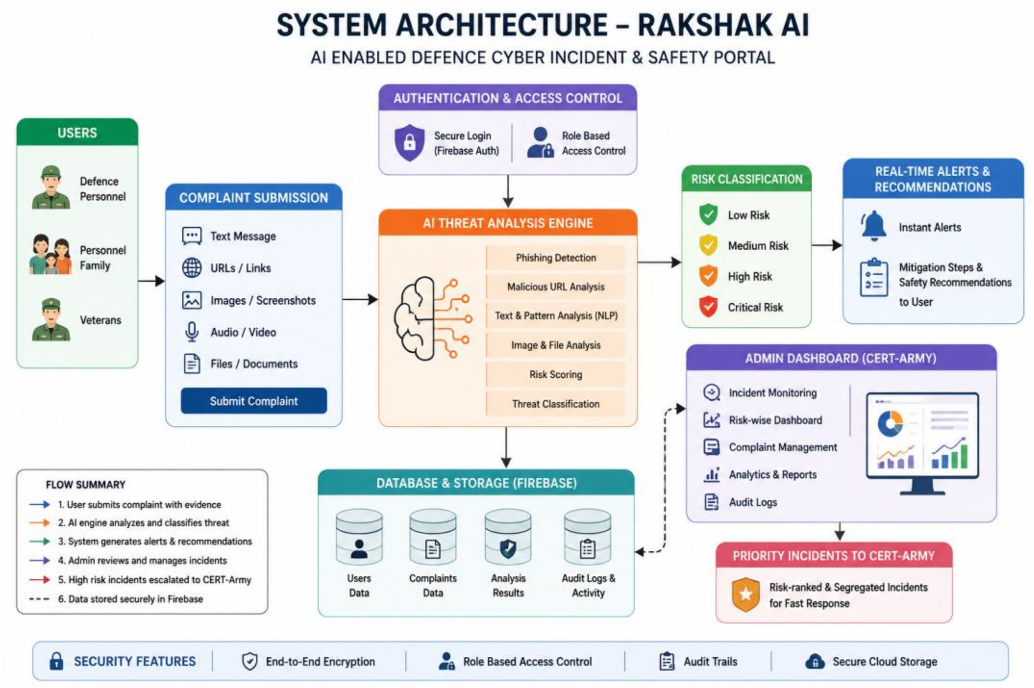

Figure 1. Rakshak AI System Architecture Overview

5.1 System Interface & Implementation

5.2 Workflow and Threat Analysis Diagram

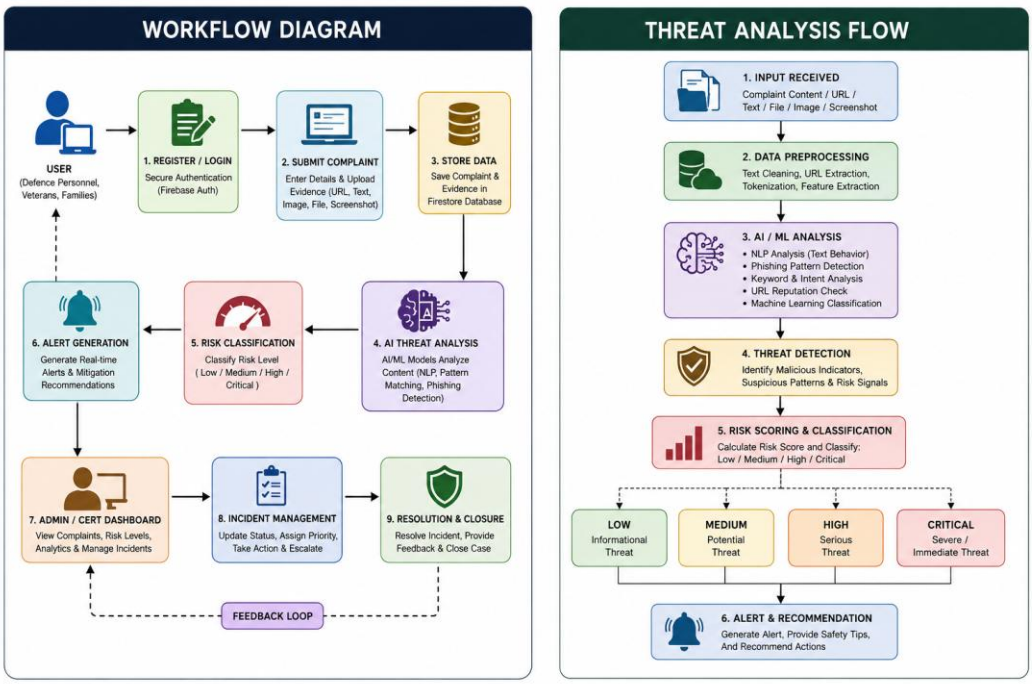

Figure 2. Rakshak AI Workflow and Threat Analysis Diagram

5.3 Workflow Screenshots

The following screenshots illustrate the key user workflows, interface screens, and operational modules of the Rakshak AI portal as observed during implementation and testing.

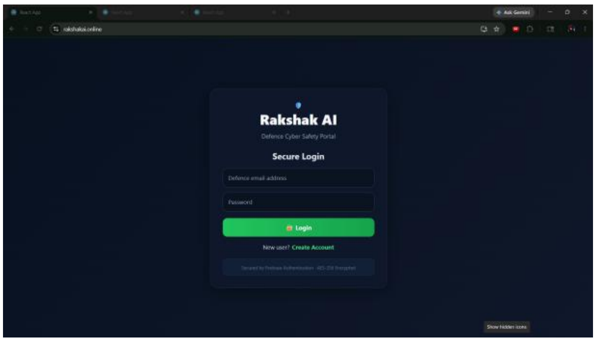

Figure 3. Rakshak AI Secure Login Interface

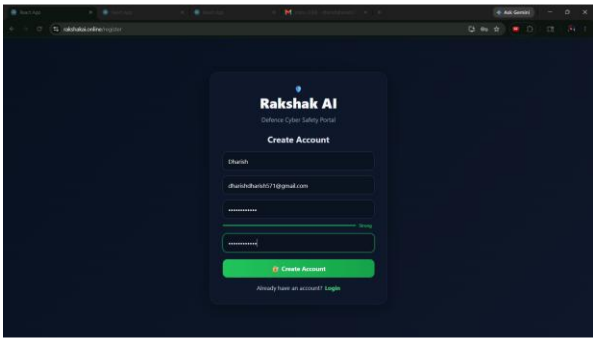

Figure 4. User Registration and Account Creation

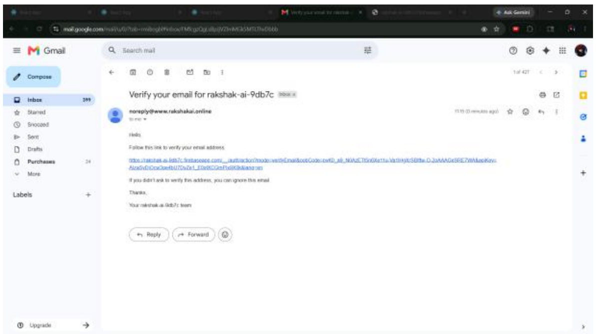

Figure 5. Automated Email Verification System

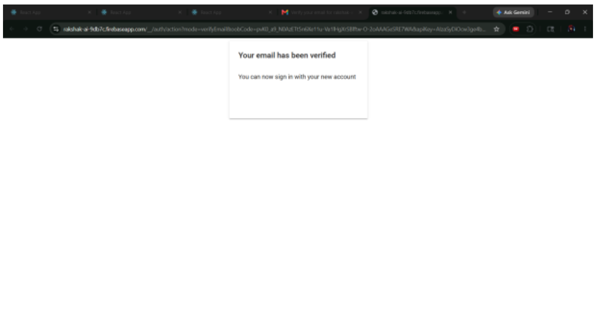

Figure 6. Successful Email Verification Confirmation

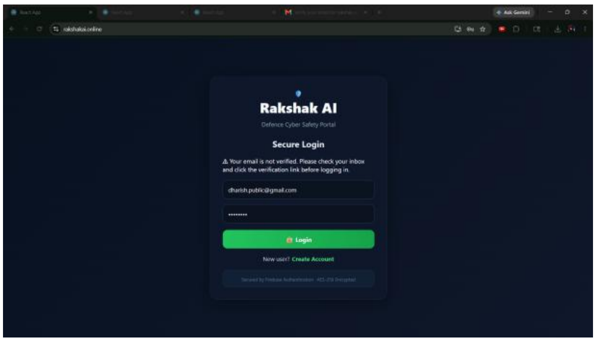

Figure 7. Unverified User Access Restriction

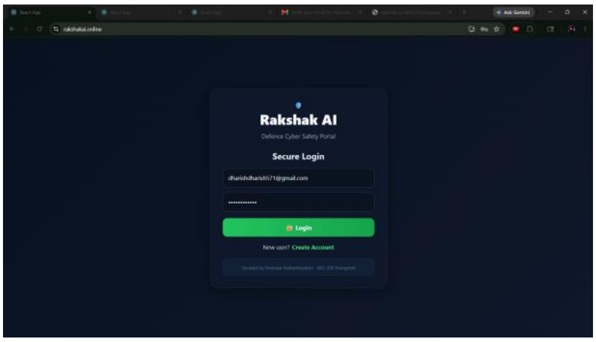

Figure 8. Authenticated User Login Portal

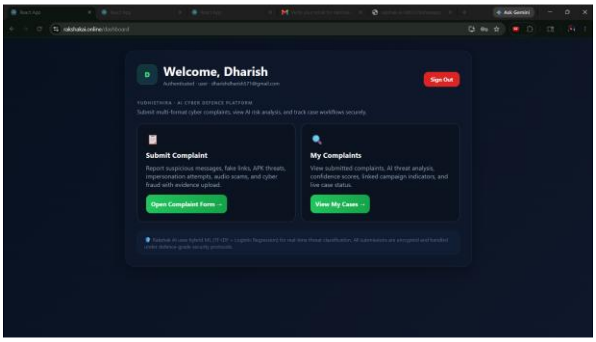

Figure 9. User Cyber Complaint Dashboard

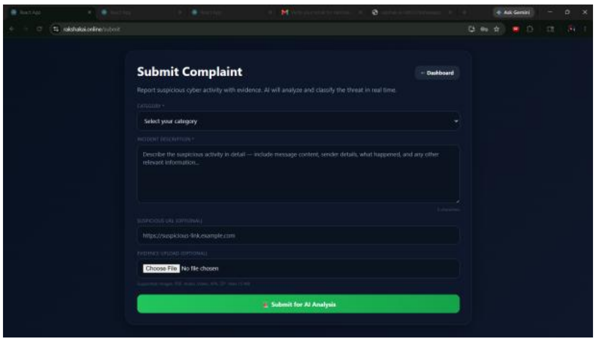

Figure 10. Cyber Complaint Submission Form

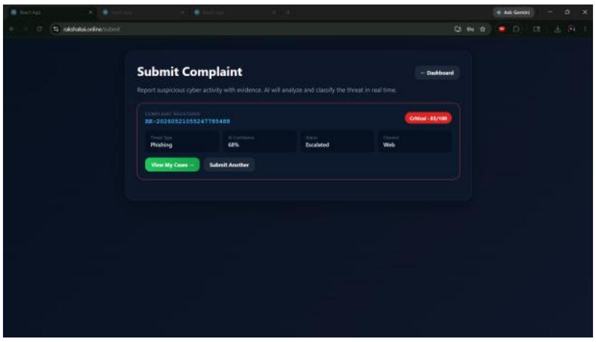

Figure 11. Complaint Submission with Suspicious Evidence

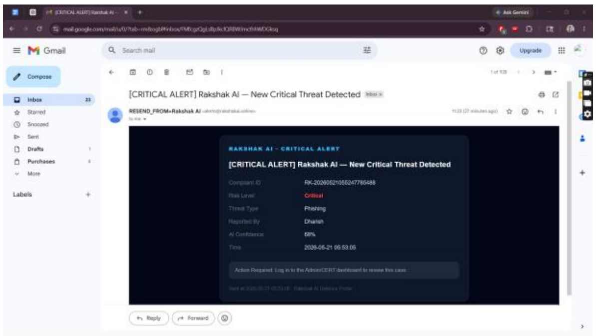

Figure 12. AI-Based Threat Classification and Risk Scoring

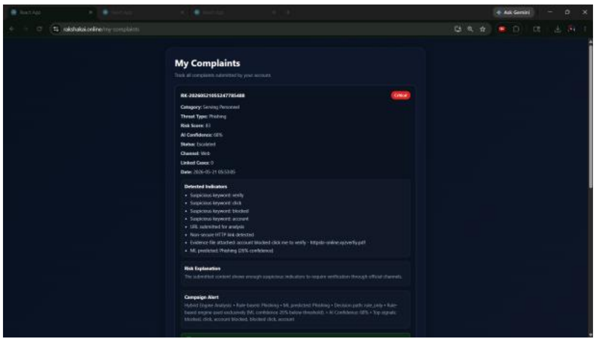

Figure 13. Automated Critical Threat Email Escalation to CERT

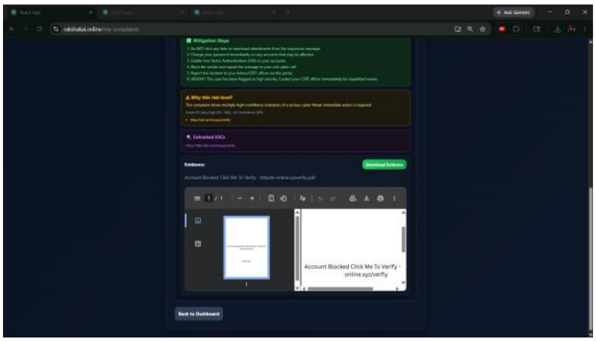

Figure 14. Threat Analysis Dashboard with Evidence Intelligence

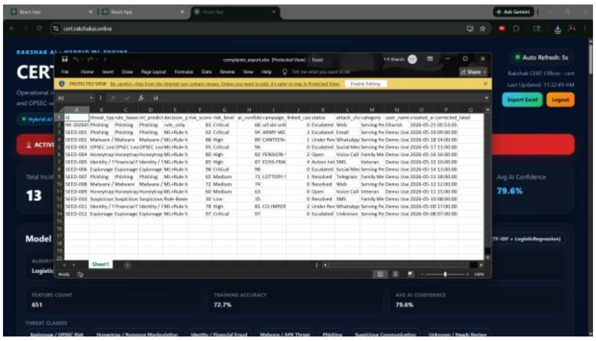

Figure 15. CERT Intelligence Dashboard with Export Capability

Figure 16. Threat Heatmap and Attack Pattern Analytics

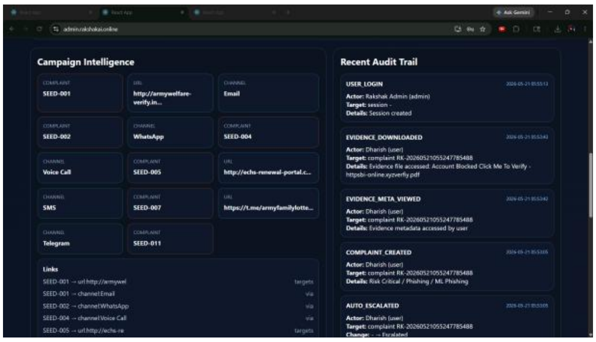

Figure 17. Administrative Audit Trail and Activity Monitoring

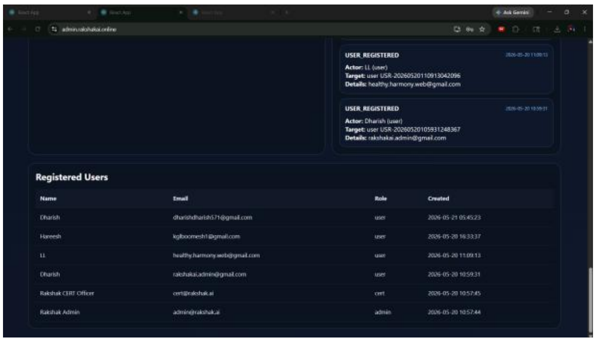

Figure 18. Registered User and Role Management Panel

6. TECHNOLOGY USED

The Rakshak AI system is developed using modern frontend technologies, AI-assisted cyber threat analysis techniques, secure authentication mechanisms, and administrative monitoring tools to provide efficient cyber incident management for defence communities.

Technology	Purpose
React.js	Frontend User Interface
Tailwind CSS	Responsive UI Design
FastAPI	Backend API Services
Firebase Authentication	Secure Login & Verification
Firestore Database	Complaint & Audit Storage
TF-IDF Vectorizer	Text Feature Extraction
Logistic Regression	Threat Classification
NLP	Scam Message Analysis
RBAC	Access Control
Gmail SMTP / Email API	CERT Alert Notifications

Technology Stack Details

Frontend Technologies

- HTML5
- CSS3
- JavaScript
- Tailwind CSS
- Vite

Backend Technologies

- Python
- FastAPI
- REST API Architecture

Database and Authentication

- Firebase Authentication
- Firestore Database
- Session-Based Authentication

Artificial Intelligence and Threat Analysis

- Artificial Intelligence (AI)
- Machine Learning (ML)
- Natural Language Processing (NLP)
- Phishing URL Detection
- Scam Message Analysis
- Risk Classification Algorithms
- Pattern and Keyword Analysis

Security and Monitoring Features

- Role-Based Access Control (RBAC)
- Secure Login System
- Audit Logs
- Incident Monitoring
- Complaint Tracking

- Risk-wise Threat Classification

Development and Collaboration Tools

- Visual Studio Code
- GitHub
- npm Package Manager
- ChatGPT
- Claude AI
- Antigravity AI

System Modules

- Complaint Submission Portal
- AI Threat Analysis Engine
- Real-Time Alert System
- Administrative Dashboard
- Analytics and Reporting Module

These technologies collectively help Rakshak AI provide secure complaint handling, intelligent cyber threat detection, real-time risk analysis, centralized monitoring, and improved digital defence preparedness for defence personnel and their families.

7. METHODOLOGY

The methodology of Rakshak AI focuses on secure cyber complaint handling, intelligent threat analysis, and efficient incident management using AI-assisted technologies. The system follows a structured workflow to detect, classify, and manage cyber threats targeting defence personnel, veterans, and their families.

Step 1: User Authentication

Users securely access the portal through a login and authentication system before submitting complaints or viewing incident details.

Step 2: Complaint Submission

Users submit cyber complaints along with digital evidence such as suspicious URLs, scam messages, screenshots, images, or files through the web portal.

Step 3: Data Processing and Threat Analysis

The submitted data is processed by the AI Threat Analysis Engine, where Artificial Intelligence, Machine Learning, and Natural Language Processing (NLP) techniques analyze the content for phishing indicators, malicious links, scam patterns, and suspicious activities.

Step 4: Risk Classification

Based on the analysis results, the system classifies incidents into multiple severity levels such as Low, Medium, High, and Critical using risk scoring and threat classification algorithms.

Step 5: Alert and Recommendation Generation

The system generates real-time alerts and provides cyber safety recommendations and mitigation steps to help users respond effectively to detected threats.

Step 6: Critical Threat Escalation

If a complaint is classified under the Critical severity category, the system automatically triggers an escalation process and sends real-time email notifications to CERT administrators for immediate incident review and response.

Step 7: Database Storage and Monitoring

All complaint details, threat analysis results, and activity logs are securely stored in the database for monitoring, tracking, and future analysis.

Step 8: Administrative Management

Authorized administrators access the dashboard to monitor incidents, view analytics, manage complaints, maintain audit records, and prioritize high-risk cyber threats for efficient handling and response management.

This methodology enables Rakshak AI to provide secure cyber complaint management, intelligent threat detection, real-time risk analysis, and centralized cyber incident monitoring for defence communities.

8. AI THREAT DETECTION ENGINE

The AI Threat Detection Engine is the core component of the Rakshak AI system responsible for analyzing cyber complaints and identifying potential security threats. The engine uses Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) techniques to detect suspicious activities, phishing attempts, malicious URLs, scam messages, and social engineering patterns.

When a user submits a complaint, the system processes the provided digital evidence such as URLs, text messages, screenshots, images, and files. The AI engine performs pattern analysis, keyword detection, and threat evaluation to identify indicators of cyberattacks or fraudulent activities. Suspicious URLs are analyzed for phishing behavior, while text-based content is checked for scam-related keywords, impersonation attempts, and malicious intent.

After completing the analysis, the engine generates a risk score and classifies incidents into Low, Medium, High, or Critical severity levels. Based on the identified threat level, the system provides real-time alerts and recommended mitigation steps to help users respond effectively to cyber threats. For incidents categorized as Critical, the AI engine also activates an automated escalation mechanism that delivers real-time email alerts to CERT administrators along with threat intelligence details and complaint metadata.

The AI Threat Detection Engine improves cyber incident analysis speed, enhances threat visibility, and supports intelligent decision-making for efficient cyber safety management within defence communities.

8.1 AI/ML Model Implementation

Rakshak AI uses a Hybrid AI-Based Threat Detection Architecture combining Machine Learning (ML), Natural Language Processing (NLP), and Rule-Based Threat Intelligence techniques for cyber threat analysis and risk classification. The system analyzes suspicious complaint text, phishing URLs, scam messages, impersonation attempts, uploaded evidence, and suspicious communication patterns using a multi-layer threat evaluation mechanism.

1. TF-IDF Vectorization

The system uses the Term Frequency–Inverse Document Frequency (TF-IDF) Vectorizer technique to convert suspicious complaint text and cyber threat content into numerical feature vectors suitable for machine learning analysis.

$$\text{TF-IDF}(t,d) = \text{TF}(t,d) \times \log(N / \text{DF}(t))$$

Where: TF = Term Frequency | DF = Document Frequency | N = Total Number of Documents

TF-IDF helps identify important phishing and scam-related keywords and was selected because it is lightweight, performs well in phishing/scam text classification, supports fast real-time inference, integrates effectively with supervised ML classifiers, and reduces computational overhead.

2. Machine Learning Threat Classification

Rakshak AI uses a supervised Machine Learning classification model for cyber threat prediction and incident categorization. The ML classifier analyzes vectorized complaint data and predicts multiple cyber threat categories including Phishing, Malware/APK Threat, Identity and Financial Fraud, Honeytrap/Romance Manipulation, Espionage/OPSEC Risk, Suspicious Communication, and Unknown/Needs Review.

3. Logistic Regression Classifier

The proposed system uses a Logistic Regression classifier integrated with TF-IDF vectorization for cyber threat classification. Logistic Regression was selected because it provides fast prediction speed, performs efficiently on NLP text classification tasks, supports real-time deployment, is computationally lightweight, provides stable classification performance, and is explainable and suitable for cybersecurity workflows.

4. Natural Language Processing (NLP)

Natural Language Processing techniques are used to perform keyword extraction, suspicious phrase detection, intent analysis, scam pattern recognition, social engineering detection, and threat sentence evaluation. The NLP engine helps detect malicious communication patterns and phishing behaviour from user-submitted complaints and suspicious messages.

5. Rule-Based Threat Intelligence Engine

A Rule-Based Threat Intelligence Engine is integrated with the ML model to improve threat detection reliability. The rule engine evaluates suspicious keywords, phishing URLs, defence impersonation attempts, fraud indicators, malicious APK references, high-risk communication behaviour, and critical cyber attack patterns. Rule-based scoring improves detection accuracy in situations where machine learning confidence scores are low or uncertain.

6. Hybrid AI Threat Classification

Rakshak AI follows a Hybrid Threat Detection Approach where TF-IDF performs text feature extraction, Logistic Regression performs threat prediction, NLP performs semantic and keyword analysis, and Rule-Based Intelligence validates critical risk indicators. The combined output is used for threat categorization, risk scoring, confidence estimation, automated CERT escalation, and real-time alert generation. This hybrid approach improves system reliability, operational efficiency, and real-time cyber threat monitoring capability.

7. AI Model Workflow

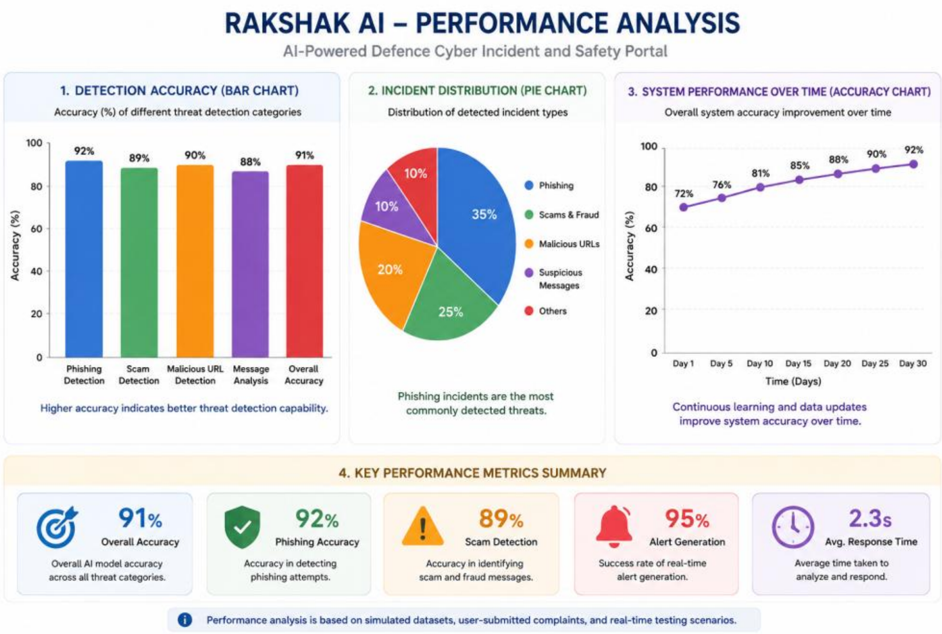

Figure 19. AI/ML Model Workflow – Hybrid Threat Detection Architecture

8. Advantages of the Proposed Model

- Fast real-time cyber threat analysis
- Lightweight deployment architecture
- Low computational overhead
- High phishing detection efficiency
- Improved explainability of predictions

- Automated threat prioritization
- Reliable hybrid risk classification
- Suitable for defence cyber safety operations
- Scalable for future AI model integration

8.2 Dataset Collection and Preparation

The Rakshak AI system was evaluated using a hybrid cyber threat dataset consisting of seeded threat records, simulated phishing campaigns, synthetic cyber fraud samples, and user-generated testing data. The dataset was prepared to simulate realistic defence-oriented cyber complaint scenarios including phishing attacks, malicious URLs, impersonation attempts, APK threats, social engineering campaigns, and suspicious communication patterns.

Dataset Type	Count / Description
Seed Threat Records	10+ pre-configured cyber threat categories
Simulated Phishing Cases	15+ synthetic attack scenarios
User Test Accounts	4 accounts for multi-user evaluation
Threat Categories	6 distinct threat categories
Uploaded Evidence Samples	10+ evidence files tested

Dataset Repository and Research Dataset Availability

To support reproducible cybersecurity research and AI-based threat evaluation, the Rakshak AI project also includes a dedicated public research dataset repository containing seeded cyber threat records, phishing URLs, suspicious communication samples, synthetic fraud scenarios, APK threat samples, and AI testing datasets. The repository was structured to simulate realistic defence-oriented cyber incident scenarios for system evaluation, threat classification testing, and AI model validation.

Dataset Repository:

https://github.com/boomeshh/rakshak-ai-research-dataset

The dataset repository supports future research extension, AI model benchmarking, phishing detection experiments, and cybersecurity awareness analysis related to defence-focused cyber threat intelligence systems.

9. DATABASE AND SECURITY ARCHITECTURE

The Database and Security Architecture of Rakshak AI is designed to ensure secure storage, controlled access, and efficient management of cyber complaint data. The system uses Firebase services and secure authentication mechanisms to protect sensitive user information and maintain reliable cyber incident monitoring.

The database layer is responsible for storing user details, cyber complaints, uploaded digital evidence, threat analysis results, risk classification data, and administrative activity records. Complaint information submitted through the portal is securely maintained for monitoring, tracking, and future analysis purposes.

To provide secure system access, Firebase Authentication is used for user login and identity verification. Only authenticated users and authorized administrators can access specific system functionalities. The portal also supports Role-Based Access Control (RBAC), where different access permissions are assigned for users and administrators to improve security and operational control.

The security architecture includes secure login validation, protected complaint handling, audit logging, and activity monitoring to maintain transparency and accountability within the system. Administrative dashboards are restricted to authorized personnel for managing incidents, viewing analytics, and monitoring high-risk cyber threats.

10. TEST CASES AND RESULTS

The Rakshak AI system was tested using multiple cyber threat scenarios to evaluate the accuracy, reliability, response speed, and effectiveness of the AI-powered threat detection and complaint management system. Various forms of suspicious inputs including phishing URLs, scam messages, fake military recruitment links, malicious files, and impersonation attempts were analyzed through the portal.

Table 1: Test Cases and Results

Test ID	Test Scenario	Input Type	Expected Output	Result
TC-01	User Login Authentication	Email & Password	Successful secure login	Pass
TC-02	Complaint Submission	Text Complaint	Complaint stored successfully	Pass
TC-03	Phishing URL Detection	Suspicious URL	High Risk Alert Generated	Pass
TC-04	Scam SMS Detection	SMS Message	Scam Warning Displayed	Pass

Test ID	Test Scenario	Input Type	Expected Output	Result
TC-05	Malware File Upload Detection	File Upload	Malicious File Warning	Pass
TC-06	Honeytrap Message Analysis	Chat Screenshot	Suspicious Behaviour Identified	Pass
TC-07	Risk Classification Engine	Multiple Inputs	Correct Risk Level Assigned	Pass
TC-08	CERT Dashboard Complaint Visibility	Complaint Data	Displayed in Admin Dashboard	Pass
TC-09	Role-Based Access Control	Admin/User Roles	Unauthorized Access Restricted	Pass
TC-10	Real-Time Alert Notification	Threat Detection	Immediate Alert Generated	Pass
TC-11	Audit Log Generation	User Activities	Activity Logged Successfully	Pass
TC-12	Safe URL Validation	Legitimate URL	Classified as Safe	Moderate
TC-13	Image Evidence Upload	Image File	Successfully Stored and Processed	Pass
TC-14	Threat Severity Prioritization	Mixed Complaints	Priority Queue Generated	Pass
TC-15	Dashboard Analytics Visualization	Threat Data	Graphical Analytics Displayed	Pass
TC-16	Critical Threat Email Escalation	Critical Risk Complaint	Automatic CERT Alert Email Generated	Pass

Results Analysis

The experimental results demonstrate that Rakshak AI effectively detects, analyzes, and classifies cyber threats with high reliability and stable system performance. The AI-assisted threat analysis engine successfully identified phishing URLs, scam SMS messages, suspicious communication patterns, malicious file uploads, honeytrap-related activities, and possible social engineering indicators.

The system successfully validated secure user authentication, complaint submission, image and file evidence handling, dashboard analytics visualization, audit log generation, and real-time alert notification functionalities. The automated Critical Threat Email Escalation mechanism generated real-time CERT alert emails whenever complaints were classified under the Critical risk category, improving rapid incident response and threat prioritization capabilities.

During local testing and simulated cyber threat analysis scenarios, the system achieved an estimated phishing and scam detection accuracy of approximately 90–92%. The average complaint processing and alert generation time remained below two seconds for standard test inputs.

11. PERFORMANCE ANALYSIS

The performance of the Rakshak AI system was evaluated using simulated phishing datasets, synthetic cyber complaint samples, and real-time testing scenarios. The AI-assisted threat detection engine demonstrated strong classification capability across phishing detection, scam identification, malicious URL analysis, and automated alert generation workflows.

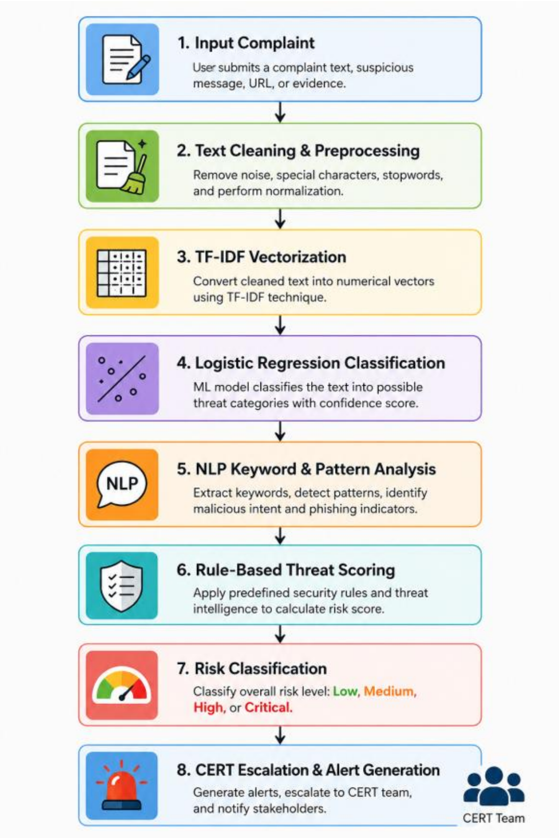

Figure 20. Performance Analysis and Detection Accuracy Metrics of Rakshak AI

Table 2: Performance Analysis Metrics

Performance Parameter	Observed Result
Phishing Detection Accuracy	90–92% (Estimated)
Scam Message Detection Accuracy	89–91%
Average Threat Analysis Time	< 2 Seconds
Complaint Submission Success Rate	98%
Dashboard Response Time	< 2 Seconds
Risk Classification Efficiency	High
Real-Time Alert Generation	Successful
Multi-Format Evidence Handling	Supported
Database Storage Reliability	Stable

Performance Parameter	Observed Result
Role-Based Access Control Efficiency	Successfully Enforced
Audit Log Generation Accuracy	100%
Concurrent Complaint Handling Capability	Moderate
System Scalability	Cloud Deployment Ready
User Authentication Reliability	Secure and Stable
Critical Alert Email Delivery	Successful

Performance Discussion

The performance evaluation results indicate that Rakshak AI performs efficiently in detecting, analyzing, and classifying cyber threats related to defence communities. The system demonstrated stable performance while processing multiple complaint submissions, monitoring dashboard activities, and generating real-time threat alerts. Firebase Authentication and Firestore Database integration provided secure user authentication, controlled access management, and reliable storage of complaint information, forensic evidence, and threat analysis records.

Although the current prototype has been tested primarily in a local development environment, the modular architecture and scalable system design support future deployment on cloud-based platforms such as Firebase Hosting, AWS, Vercel, and Render for improved accessibility, scalability, and production-level deployment.

12. ADVANTAGES OF THE SYSTEM

The Rakshak AI system offers several advantages by combining Artificial Intelligence, secure authentication mechanisms, and cyber threat analysis capabilities into a unified defence-oriented platform.

- **AI-Based Threat Detection:** The system uses Artificial Intelligence techniques to identify phishing links, scam messages, suspicious communication patterns, and possible cyber threats with improved efficiency.
- **Faster Complaint Analysis:** Rakshak AI automatically processes and classifies complaints based on risk severity, reducing manual effort and improving incident response speed.
- **Defence-Oriented Cyber Safety Platform:** Unlike general public cybercrime portals, the system is specifically designed for defence communities, enabling better prioritization of sensitive cyber incidents.
- **Real-Time Threat Alerts:** The portal generates immediate warnings and safety recommendations whenever suspicious or malicious content is detected.
- **Multi-Format Complaint Support:** Users can submit various forms of cyber evidence including URLs, text messages, screenshots, images, and files for analysis.
- **Secure Authentication System:** Firebase Authentication and role-based access mechanisms help ensure secure user login and controlled access to sensitive complaint information.

- **Automated CERT Escalation:** The system automatically escalates Critical-level cyber complaints and sends real-time email alerts to CERT administrators, enabling faster incident response.
- **User-Friendly Interface:** The frontend developed using React provides an interactive and responsive user experience for complaint submission and dashboard monitoring.
- **Centralized Complaint Management:** The administrative dashboard enables organized complaint tracking, monitoring, and status management in a single platform.
- **Audit Logging and Monitoring:** The system maintains activity logs and monitoring records to improve accountability and support future investigation processes.
- **Scalable System Architecture:** The modular architecture allows future deployment on cloud platforms and supports integration of advanced cybersecurity features.

13. FUTURE ENHANCEMENT CAPABILITY

Additional functionalities such as deepfake detection, multilingual support, mobile application integration, and advanced AI threat intelligence can be incorporated in future versions of the system.

13.1 Limitations

- Current system tested mainly in local development environment
- AI-assisted threat analysis evaluated using simulated and seeded datasets
- Large-scale real-world deployment not yet performed
- Limited multilingual phishing analysis support

The proposed system demonstrates the potential of AI-assisted cyber defence platforms in improving proactive cyber threat monitoring and incident response for defence communities.

13.2 Real World Applications

The Rakshak AI platform can be extended and deployed across multiple real-world cybersecurity environments including defence cyber safety monitoring, phishing incident management, military communication protection, cyber awareness platforms, educational cybersecurity training systems, and government cyber complaint management infrastructures.

The proposed architecture may also support future integration with CERT-In workflows, cyber forensic systems, SIEM dashboards, threat intelligence platforms, and cloud-based cyber monitoring environments for large-scale cyber threat management and proactive defence preparedness.

POTENTIAL REAL-WORLD IMPACT

- Defence cyber awareness and preparedness
- AI-assisted phishing detection and mitigation
- Centralized cyber complaint intelligence handling
- Faster CERT escalation and threat response
- Cybersecurity training and educational support

- Defence-focused cyber threat monitoring
- Scalable AI-assisted cyber incident management

14. CONCLUSION

In this paper, Rakshak AI, an AI-powered Defence Cyber Incident and Safety Portal, was successfully designed and developed to address the increasing cyber threats targeting defence personnel, veterans, and their families. The proposed system provides a dedicated platform for secure cyber complaint registration, intelligent threat analysis, risk classification, and centralized monitoring of cyber incidents.

The integration of Artificial Intelligence techniques enabled the system to identify phishing attacks, scam messages, suspicious URLs, and possible social engineering threats with improved efficiency. The implementation of real-time alert generation, role-based access control, audit logging, and complaint prioritization further enhanced the security and reliability of the platform. The integration of automated CERT escalation and email-based critical threat notification mechanisms further improved the responsiveness and operational effectiveness of the proposed system.

The system demonstrated stable performance during local testing and successfully supported multiple complaint handling operations, secure authentication processes, and AI-assisted threat analysis functionalities. The modular architecture and responsive frontend design also provide flexibility for future deployment and scalability enhancements.

Rakshak AI contributes toward strengthening cyber safety awareness and proactive threat management for defence communities by combining intelligent automation, secure infrastructure, and user-friendly cyber incident reporting mechanisms. Future improvements such as advanced AI models, cloud deployment, mobile application support, multilingual assistance, and deepfake detection capabilities can further enhance the effectiveness and real-world applicability of the proposed system.

The proposed system demonstrates the feasibility of integrating AI-assisted cyber threat intelligence with defence-oriented cyber incident management frameworks for future scalable national cybersecurity applications.

15. SOURCE CODE AVAILABILITY

The Rakshak AI prototype system, seeded threat dataset repository, and supporting research artifacts were developed for academic and cybersecurity research purposes under YUDHISTRA 2026. Selected implementation modules, seeded datasets, and testing resources are maintained through GitHub repositories for research validation and future enhancement support.

Research Dataset Repository:

https://github.com/boomeshh/rakshak-ai-research-dataset

Prototype / Source Code Repository:

Private research repository available for academic evaluation upon request.

ETHICAL DISCLAIMER

The Rakshak AI project was developed strictly for academic research, cybersecurity awareness, cyber threat simulation, and defence-oriented incident management research purposes. All phishing samples, seeded threat records, suspicious communication examples, and synthetic datasets used during experimentation were generated or collected exclusively for controlled testing and educational evaluation.

The proposed system does not promote unauthorized cyber activity, offensive cyber operations, or malicious exploitation. Any resemblance to real-world incidents is purely for research simulation and cybersecurity evaluation purposes.

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APPENDIX – PROJECT RESOURCES & DEMO LINKS

A.1 Research Dataset Repository

https://github.com/boomeshh/rakshak-ai-research-dataset

Contains seeded cyber threat records, phishing URLs, synthetic fraud scenarios, and AI testing datasets for academic evaluation.

A.2 Prototype Repository

Private research repository available for academic evaluation upon request.

A.3 GitHub QR Code

A.4 Demo QR Code

A.5 Architecture Diagram Reference

Refer to Figure 1 (System Architecture Overview) and Figure 19 (AI/ML Model Workflow) in Section 5 and Section 8 respectively for complete architectural diagrams of the Rakshak AI system.

A.6 System Demo Screenshot Placeholder